

# Islam Tarek Moawad

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## Summary

Electronics and Communications Engineering student specializing in PCB Design and hardware development. Experienced in designing multi-layer PCBs using KiCad, with hands-on knowledge of schematic design, component selection, PCB layout, routing, power distribution, and design for manufacturing. Familiar with high-speed and RF-aware PCB design concepts, simulation workflows, and preparing designs for real-world fabrication. Developed and documented several hardware projects, including an RP2040 development board and a mechanical vibration analysis system. Strong foundation in electronics, embedded systems, C/C++, Python, MATLAB, and technical documentation, with a focus on building practical and manufacturable electronic systems.

## EXPERIENCE

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|--------------------------------------------------------------|----------------------------------------|
| <b>IEEE-HSB ComSoc</b><br><i>Analog IC Design instructor</i> | Oct. 2025 – Present<br>Dokki, Giza     |
| <b>Marine scout family @ FEHU</b><br><i>Member</i>           | Oct. 2024 – May. 2025<br>Cairo, Helwan |
| <b>Batch leader</b><br><i>Batch 61 FEHU leader</i>           | oct 2023 – Feb 2025<br>Cairo, Helwan   |
| <b>Atlantis</b><br><i>Hardware and Head</i>                  | Sep 2025 – now<br>Cairo, Helwan        |

## EDUCATION

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|----------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Faculty of Engineering in Helwan, Capital University</b><br><i>Junior Electronics and Communications Engineer</i> | 2023– 2028<br>Cairo, Helwan |
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## SKILLS

**Soft Skills** : Problem Solving, Analytical Thinking, Self-Learning, Teamwork, Communication, Time Management, Leadership, Attention to Detail

**Programming languages** : Python, C, C#, MATLAB,  $\LaTeX$

**Tools** : KICAD, FreeCad, Atmel Studio, Arduino IDE, ESP32-s3 DevKit, Visual Studio Code, Proteus, Multisim, LTspice, LabView, MATLAB, Logic Friday, VMware, Virtual Machine Manager(QEMU/KVM), TeXstudio, Unity Engine, GitHub, Git, Super Productivity, Obsidian

**Operating Systems I'm familiar with** : Windows, Debian, Ubuntu, Kali Linux, Arch Linux

## Certifications

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|-------------------------------------------------------------------------|----------|
| <b>Microcontrollers systems, NTI summer training program</b>            | 2025     |
| <b>MATE ROV EGYPT</b><br><i>For Participating in the MATE ROV EGYPT</i> | May 2026 |
| <b>High-Speed design with KiCad , Udemy</b>                             | Sep 2026 |

## PROJECTS

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### Google Drive Automation For Academic Materials

- A Python-based automation system that relies on Google services (Google Cloud, Google Sheets API, Google Drive API, Google Forms) to organize summaries and academic materials for my Batch
- It took two weeks to build:
  - \* 1st week: Learning Python and Google Cloud
  - \* 2nd week: Building the system
- 200 files were uploaded using this system, and the files shared on it remained for newer batches.

### ECG R-Pulse Detector

- This is the project I'm most proud of.
- This project was required for the Analog Electronics course.
- No specific ideas were given, so I chose to do the most challenging possible project.
- The project duration was two months and I reached the correct electronic circuit three days before the deadline.

### Metro Ticket Machine Algorithmic State Machine

- This was a project in the Logic Circuits course for a team of 10 people.
- I led the team and designed the most complex Algorithmic State Machine in the batch myself in half an hour.

### Large Smart Home System With 17 Atmega32

- this was my graduation project of the NTI Microcontrollers systems internship.
- It was a stressful project, more than a difficult one; I learned patience from it.

### Analog Implementation Of Neural Networks

- This was one of the first ideas that came to me after learning the basics of Neural Networks.
- The project was carried out using a set of operational amplifiers, and it was one of the most stressful projects I've ever undertaken.

### ROV Control And Communication System

- It was during my participation in the MATE Egypt ROV competition as a hardware head.

### ROV PCB Design

- Designed PCB hardware for an underwater Remotely Operated Vehicle (ROV), focusing on integrating the electronic subsystems into a practical board design.
- Developed the schematic, PCB layout, component placement, and routing using KiCad.
- Worked on the hardware architecture required for an embedded underwater robotic platform.

### RP2040 Development Board

- Designed a custom four-layer RP2040 development board in KiCad based on the official hardware design reference.
- Developed the complete schematic, component and footprint selection, PCB layout, routing, and supporting hardware circuitry.
- Performed 3D inspection and design verification of the board, including layer-by-layer visualization and manufacturing documentation.

### Mechanical Vibrations Analyzer

- Designed a custom six-layer PCB for an embedded mechanical vibration monitoring system.
- Integrated vibration sensing, embedded signal processing, and a display for visualizing machine vibrations as a frequency-versus-time spectrogram.
- Designed a custom mechanical enclosure in FreeCAD and developed the system for direct mounting onto machinery.